

HAMILTON, TX, USA

WMO: 722231

Lat: 31.666N

Lon: 98.149W

Elev: 396

StdP: 96.66

Time Zone: -6.00 (NAC)

Period: 08-19

WBAN: 00357

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.9	-2.5	-12.6	1.3	1.2	-10.2	1.7	3.3	13.2	14.0	11.8	12.3	4.5	10	0.502

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.5	38.7	21.5	37.4	21.8	36.2	22.1	24.9	29.8	24.4	29.6	24.0	29.5	4.2	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.1	19.9	25.0	23.6	19.3	24.9	22.9	18.5	24.8	77.9	29.7	75.8	29.0	74.4	29.3	26.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.1	9.8	8.8	-8.9	40.1	1.3	2.0	-9.9	41.5	-10.7	42.7	-11.4	43.8	-12.4	45.3		
			-9.8	25.6	1.3	0.7	-10.7	26.1	-11.5	26.5	-12.2	26.9	-13.1	27.5		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	19.4	8.8	11.1	15.0	18.9	23.0	27.6	29.2	29.3	25.4	19.9	14.2	9.6
	DBStd	8.49	5.34	6.28	5.27	4.23	3.55	2.20	1.89	2.58	3.28	4.73	5.21	5.47
	HDD10.0	258	85	57	18	1	0	0	0	0	0	2	21	75
	HDD18.3	1138	296	211	128	43	7	0	0	0	1	39	139	274
	CDD10.0	3684	49	87	172	268	404	527	594	599	463	311	147	63
	CDD18.3	1522	2	8	24	61	153	277	336	341	214	88	16	3
	CDH23.3	17469	31	68	221	562	1400	3208	4311	4434	2245	834	121	34
	CDH26.7	8944	4	12	48	173	547	1672	2464	2587	1109	299	23	4
Wind	WSAvg	4.3	4.4	4.9	5.1	5.3	4.7	4.3	3.8	3.5	3.3	4.0	4.2	4.3
Precipitation	PrecAvg	852	48	66	70	75	95	102	57	68	73	88	58	49
	PrecMax	1256	112	203	197	189	242	280	155	227	149	178	209	230
	PrecMin	504	9	0	3	12	21	25	0	0	12	14	1	5
	PrecStd	202	26	48	42	46	55	67	44	61	41	51	54	49
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.8	28.0	30.1	33.7	34.7	38.0	40.4	40.3	37.9	34.4	29.4	27.1
		MCWB	14.5	15.1	16.7	19.2	22.1	22.1	21.5	21.3	20.2	20.2	19.4	16.2
	2%	DB	23.6	25.2	27.9	30.5	32.9	36.7	38.1	38.9	36.0	31.6	26.5	23.7
		MCWB	13.5	16.1	16.9	19.1	21.5	22.2	21.6	21.7	21.3	20.4	17.4	15.3
	5%	DB	20.5	22.9	26.0	28.5	31.5	35.2	37.0	37.7	34.0	30.0	24.2	21.0
		MCWB	12.9	14.9	16.5	18.8	21.3	22.4	21.9	22.0	21.2	19.8	17.1	14.5
	10%	DB	17.7	20.9	23.7	26.7	30.0	33.9	35.7	36.6	32.6	28.0	22.4	18.6
		MCWB	11.1	14.6	15.6	18.3	21.0	22.5	22.3	22.2	21.0	18.4	16.1	13.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.1	20.2	20.6	22.9	24.8	25.6	25.4	25.1	25.0	23.9	22.2	20.0
		MCDB	21.2	22.7	24.2	27.1	29.0	31.2	30.2	30.0	28.5	26.9	25.0	21.7
	2%	WB	16.2	19.0	19.5	21.7	23.9	24.7	24.8	24.5	24.2	23.1	20.6	18.3
		MCDB	20.2	22.1	23.5	26.2	28.7	30.0	29.9	30.3	28.3	26.2	22.4	21.0
	5%	WB	14.2	17.8	18.7	20.9	23.2	24.1	24.3	24.0	23.7	22.4	19.1	16.6
		MCDB	18.0	20.6	22.8	26.0	27.9	29.4	29.8	30.4	28.1	26.1	21.3	18.7
	10%	WB	12.4	16.0	17.5	20.1	22.6	23.7	23.8	23.5	23.0	21.4	17.9	14.6
		MCDB	17.0	19.4	21.7	24.3	27.2	29.2	29.8	30.2	27.8	25.5	20.9	18.3
Mean Daily Temperature Range	5% DB	MDBR	12.5	12.2	12.9	12.6	11.4	11.9	12.7	12.5	12.1	12.6	12.3	11.7
		MCDBR	16.4	14.9	14.6	14.7	13.0	13.6	14.3	14.3	13.9	13.9	13.7	15.1
		MCWBR	8.7	8.4	6.6	6.8	4.5	2.9	3.2	3.1	3.9	5.5	6.8	8.7
	5% WB	MCDBR	13.7	12.0	11.7	11.0	10.6	11.0	11.7	11.6	10.7	10.8	10.4	11.6
		MCWBR	9.4	8.5	6.8	6.3	4.6	3.2	3.1	3.0	3.3	6.0	7.2	9.2
Clear-Sky Solar Irradiance	taub	0.305	0.315	0.341	0.380	0.413	0.414	0.425	0.416	0.389	0.356	0.328	0.310	
	taud	2.520	2.500	2.440	2.355	2.320	2.355	2.366	2.372	2.393	2.471	2.488	2.512	
	Ebn at Noon	926	948	941	911	878	872	860	866	879	888	889	900	
	Edh at Noon	85	96	110	124	130	125	123	121	114	98	88	82	
All-Sky Solar Radiation	RadAvg	3.06	3.64	4.69	5.66	5.96	6.75	6.82	6.27	5.25	4.19	3.21	2.71	
	RadStd	0.33	0.56	0.39	0.51	0.58	0.55	0.51	0.41	0.44	0.63	0.41	0.36	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page